MIL-C-44116
12 December 1983
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

CARROTS, DEHYDRATED, COMPRESSED, DICED OR SHREDDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers dehydrated, compressed carrots in disc form for use by the Department of Defense as an item of general issue.

1.2 Classification. The carrots shall be of the following types as specified (see 6.1).

Type I - Diced (prepared from fresh or frozen carrots)
Type II - Shredded (prepared from fresh carrots only)

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8915

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MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit, Preparation of Nonperishable Subsistence Items; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standard for Condition of Food Containers

(Application for copies should be addressed to the Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Process Products Thereof, and Certain Other Processed Food Products

U.S. Standards for Grades of Frozen Carrots

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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U.S. Standards for Grades of Topped Carrots

(Application for copies should be addressed to the Chief, Fresh Products Inspection, Fruit and Vegetable Division, Agricultral Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 No. 19th Street, Arlington, VA 22209.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milks, Including Methods of Analysis, Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

AMERICAN CHEMICAL SOCIETY

Analytical Reagents

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, N.W., Washington, DC 20002.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.4).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors and off-colors.

3.2.1 Carrots. The carrots shall be Imperator or Red Cored Chantenay or other varieties with comparable quality and suitability for dehydration. They shall be clean, sound, mature, of good typical orange color, and of good cooking quality. Frozen diced carrots conforming to U.S. Grade A (Fancy) of the United States Standard for Grades of Frozen Carrots of the latest harvest which meets the above requirements may be used provided sulfiting is accomplished subsequent to complete thawing. Fresh carrots conforming to U.S. No. 1 Grade of the United States Standard for Grades of Topped Carrots shall be used for both type I and type II.

3.2.2 Sodium sulfite or sodium acid sulfite. The sulfite(s) used shall meet the requirements of the Food Chemicals Codex (F.C.C.).

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3.3 Preparation and processing.

3.3.1 Fresh carrot preparation for type I and II). The carrots shall be washed thoroughly to remove soil and foreign material and sorted to remove any which are diseased or otherwise damaged. The sound stock shall be peeled and trimmed to remove discolored, damaged, or undesirable portions. The carrots, for type I shall be cut into nominal 3/8 inch dice and the carrots for type II shall be cut into shreds not less than 1/2 inch in length using a 3/16 inch grating plate.

3.3.2 Frozen carrot preparation for type I only). The frozen carrots shall have been previously cut into 3/8 by 3/8 by 3/8 inch dice ± 1/16 inch in any dimension.

3.3.3 Blanching. The cut carrots shall be blanched in live steam until the peroxidase in the carrots is inactivated when tested as specified in 4.5.2.2. If frozen diced carrots which have been blanched only t catalase inactivation are used for type I, the carrots shall be further blanched to inactivate peroxidase.

3.3.4 Sulfiting. The cut fresh and frozen diced carrots after thawing shall receive a sulfite treatment by such method s) as will insure a reasonably uniform distribution throughout the product and a residual sulfur dioxide content in the dehydrated product within the limitation specified in 3.4.1. The sulfite used shall conform to the Food Chemicals Codex. Contact of the wet carrots with iron, subsequent to the sulfiting operation, shall be strictly avoided.

3.3.5 Freezing. The blanched and sulfited carrots shall be quick frozen as expeditiously as possible after sulfiting and held in the frozen state until dehydrated.

3.3.6 Dehydration. The frozen product shall be freeze-dehydrated at an absolute pressure not exceeding 1.0 millimeter of mercury. At no time shall there be evidence of thawing in the product. After dehydration has been completed, the pressure in the freeze-drying chamber shall be equalized to atmospheric pressure with nitrogen and the product shall be preserved immediately as specified in 5.1, or shall be held under an atmosphere of nitrogen not more than 2.0 percent oxygen) for not more than 16 hours until packaging can be effected.

3.3.7 Conditioning. The freeze dried carrots shall be conditioned by adding moisture water spraying only, since steaming causes bleaching of the color) sufficient to make them pliable so that they will not be crushed or ground under high pressure. Note: Satisfactory compressed carrots have been produced by spraying with moisture to 9-12 percent and placing the sprayed carrots in a closed container and heating approximately 10 minutes at 150°F in order to equilibrate the moisture in the carrots.)

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3.3.8 Mechanical pressing. The moistened carrots shall be placed in a 3-5/8 ± 1/8 inch diameter mold and compressed into discs by applying adequate pressure and dwell time to yield cohesive discs which resist breakage during handling and packaging. Satisfactory results have been obtained by applying 100-250 pounds per square inch of pressure on each disc for 10-15 seconds depending upon the variety and maturity of the carrots.

3.3.9 Re-drying. The compressed carrots shall be redried within a minimum of time so that the finished product complies with 3.4.

3.4 Finished product.

3.4.1 Compressed product. The compressed dehydrated product shall be free from hay-like, scorched, oxidized (violet-like odor), rancid, musty, or other objectionable odors; shall not exceed 3.0 percent by weight of moisture; and shall contain a sulfite level, calculated as sulfur dioxide, of 500 ± 250 ppm.

3.4.2 Rehydrated product. The rehydrated type I product shall possess the color, odor, flavor and appearance typical of cooked diced frozen carrots and a texture which is tender but not mushy. The rehydrated type II product shall possess the color, odor, flavor and appearance typical of blanched shredded carrots and shall have a crisp firm texture. The rehydrated products (type I and type II) shall contain no more than 1.0 percent by weight of units damaged by pathological or insect injury and no more than 5.0 percent by weight of extraneous vegetable matter (common to the carrot plant), heat damaged units, blemished units, or otherwise damaged units.

3.4.3 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection see 4.4).
b. Quality conformance inspection see 4.5).

4.4 First article inspection. When a first article is required see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurane provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.1.2 Can tin coating weight testing. The cans shall be tested in accordance with PPP-C-96 for conformance to the tin coating weight requirement in 5.1.1.

4.5.2 In-process inspection.

4.5.2.1 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature and formulation or other valid documents shall be cause for rejection of the involved product.

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4.5.2.2 In-process testing. As each production run is initiated, a sample of the blanched carrots, prior to dehydration, shall be tested in accordance with 4.6.4. A test indicating active peoxidase enzymes shall be cause for rejection of that portion of the production from the time of previous negative result until such time as a negative test result is again attained.

4.5.3 Net weight examination. The cans of product shall be examined for net weight. Any individual sample having a net weight of less than 15.0 ounces for type I or less than 11.0 ounces for type II shall be scored as a minor defect. The lot size shall be expressed in units of cans of one type of product only. The sample unit shall be one can of product. The inspection level shall be S-4 and the acceptable quality level AQL), expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 16.0 ounces for type I or less than 12.0 ounces for type II.

4.5.4 Product examination. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in cans of one type product only. The sample unit shall be the contents of one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	Dehydrated product
151	Presence of hay-like, scorched, rancid, oxidized violet-like odor), musty, or other objectionable odors.
	Rehydrated product 3/
152	Mushy texture.
153	More than 5.0 percent, by weight, of extraneous vegetable matter common to the carrot plant, heat damaged units, blemished or otherwise damanged units.
154	More than 1.0 percent of units are damaged by pathological or insect injury.
155	Color, odor, flavor, texture, or shape not as specified.

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TABLE I. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	Type I
156	Texture not tender.
157	Color, odor, flavor or appearance not typical of cooked diced frozen carrots (see 3.4.2).
	Type II
158	Texture not firm and crunchy.
159	Color, odor, flavor or appearance not typical of blanched shredded carrots (see 3.4.2).

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ Type I dehydrated compressed carrots shall be rehydrated by adding 15.0 parts of boiling water to 1.0 part of carrots, by weight, stirring gently to break apart, bringing to a boil and simmering for 5 minutes. Type II dehydrated compressed carrots shall be rehydated by adding 10.0 parts of tap water (60°F) to 1.0 part of carrots, by weight. Break each disc into four approximately equal portions and allow to fully rehydrate (approximately 45 minutes) with occasional stirring.

4.5.5 Moisture and sulfur dioxide content testing. The finished product shall be tested as specified in 4.6.1 and 4.6.2 for conformance to the moisture and sulfur dioxide requirements in 3.4.1. The sample for testing shall be a 1-pound composite, derived from the number of cans indicated by inspection level S-2. The lot size shall be expressed in units of cans of one type product only. Results shall be reported to the nearest 0.1 percent for moisture and to the nearest 1.0 ppm for sulfur dioxide. A test failure for either moisture or sulfure dioxide content shall be case for rejection of the lot.

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4.5.6 Oxygen in headspace testing. The filled and sealed cans shall be tested as specified in 4.6.3 for conformance to the oxygen in headspace requirement in 5.1.1. The lot size shall be expressed in cans of one type of product only. The sample unit shall be one filled and sealed can. The inspection level shall be S-2 and the AOL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.7 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.5.8 Leakage and vacuum (when applicable) inspection. Inspection for leakage and vacuum shall be in accordance with PPP-C-29 (see 5.1.1).

4.5.9 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

Major:	National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible. Arrangement or number of cans not as specified.
Minor:	Other required markings missing, incorrect, or illegible. Reinforcement with nonmetallic strapping or tape is not used.

Level C shipping containers shall be examined for the marking, arrangement, and number of can defects specified above and for the closure method specified in 5.2.3.

4.5.10 Unit load inspection. Inspection of unit loads shall be in accordance with quality assurance provisions of MIL-L-35078.

4.6 Methods of inspection.

4.6.1 Moisture content test. Prepare a clean, dry Waring blender or similar mill by grinding a sample and discarding it. Grind approximately 100 grams of a well mixed sample in the prepared blender. Sieve the ground material through a U.S. Standard No. 20 sieve and collect that portion which passes through in a clean, dry, tightly closed container. Weigh two samples, approximately 2 grams each of the sieved material, into tared, dry, aluminum weighing dishes (approximately 2 or 2-1/2 inches in diameter and 3/4 inch in depth) with tight-fitting covers. Place the dishes with cocked lids in a

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vacuum oven and dry for 6 hours at 70°C under a pressure of not more than 100 mm mercury. During drying, admit to the oven a slow current of air (approximately two bubbles per second) dried by passing through concentrated sulfuric acid. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

4.6.2 Sulfur dioxide content test. Sulfur dioxide of the ground sample shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Food Additives; section: Chemical Preservatives, method: Total Sulfurous Acid, Modified Monier-Williams.

4.6.3 Oxygen content test. Oxygen content of head space gas shall be determined in accordance with the American Dry Milk Institute, Inc., publication, Bulletin 916, Standards for Grades of Dry Milks, Including Methods of Analysis: Determination of Oxygen. The result of each determination shall be reported to the nearest 0.1 percent.

4.6.4 Peroxidase activity. The test to show adequate blanching shall be made in accordance with the following methods:

a. Reagents: (Reagents shall be reagent grade, Analytical Reagent or Chemically pure as defined by the American Chemical Society in the A.C.S. Analytical Reagents or meet the requirements of the Food Chemical Codex.)

Distilled water.
0.5 percent guaiacol in 50.0 percent ethyl alcohol solution.
0.08 percent hydrogen peroxide (2.8 mL of 30.0 percent hydrogen peroxide per liter). Keep in refrigerator in dark bottle and renew solution each week or two.
Clean sand.

b. Apparatus:

Test tubes, 3/4 or 7/8 inch in diameter.
3 or 4 inch diameter funnels.
6 or 7 inch cotton milk filters.
Waring blender or similar mixer or porcelain mortar (4 to 6 inch diameter) and pestle.
50.0 mL graduated cylinder.
1.0 and 2.0 mL pipettes.
Timer or watch with second hand.
Test tube rack.
Balance which will weigh 10.0 gram samples to $\pm$ 0.1 gram (A triple-beam type balance is recommended).
If blender is used, a larger sample will permit use of less sensitive balance.

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c. Procedure, using blender:

1. Weigh out representative 100 to 200 gram sample.
2. Place in blender with 3.0 mL water to each gram of vegetable.
3. Grind for 1 minute at moderate or high speed.
4. Filter through cotton milk filter.
5. Add 2.0 mL of filtrate to 20 mL of distilled water in test tube.
6. Prepare a blank by adding 2.0 mL of filtrate to 22.0 mL of distilled water in a second test tube, mix and use as a color comparison tube. (Do not add any guaiacol or peroxide to this tube.)
7. Add 1.0 mL of 0.5 percent guaiacol solution to first test tube without mixing.
8. Add 1.0 mL of 0.08 percent hydrogen peroxide to the same tube without mixing.
9. Mix contents thoroughly by inverting and watch for development of any color differing from the blank, regardless of hue, but of sufficient intensity to show an obvious contrast. Development of color is a positive test and indicates inadequate blanching. If no color contrast develops in 3-1/2 minutes, consider the test negative and the product adequately blanched. If color develops after 3-1/2 minutes, it is to be disregarded and the test still considered negative (for further information see full report, Masure, M.P. and Campbell, H., "Rapid Estimation of Peroxidase in Vegetable Extracts - An Index of Blanching Adequacy for Frozen Vegetables." Fruit Prod. Jour. and American Food Mfr. 23:369-375, 384, Aug. 1944).

d. Procedure, using mortar:

1. Weigh representative 10.0 gram sample.
2. Place 30.0 mL distilled water in graduated cylinder.
3. Place sample in mortar with a little clean sand, add minimum amount of water from graduated cylinder to give best consistency for thorough maceration, and grind for 3 minutes. Add remainder of water from cylinder and mix.
4. Proceed as under (c) above, 4 thorugh 9 inclusive.

NOTE: Because of the limits of mortar grinding when using mortar with 10.0 gram samples, special attention should be given to assuring representative samples. Sample size can be increased to 20.0 to 30.0 grams with additional time given to grinding, or the number of units represented can be increased by halving or quartering, if the sample is uniform.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C, as specified (see 6.1).

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5.1.1 Level A. A net weight of sixteen ounces type I or twelve ounces of type II of compressed discs of the product shall be unit packed in a size 401 by 411 can. A minus tolerance of 1.0 ounce will be permitted in any individual can provided that the average net weight of cans examined in accordance with 4.5.3 is not less than 16 or 12 ounces as applicable. The can shall be an open top style, round, metal can with soldered or welded side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25-pound per base box electrolytic tin plate see 4.5.1.2), and shall be coated overall on the outside with a coating conforming to type I or when specified type III see 6.1) of TT-C-495. The can shall have a sufficient base plate weight and temper to protect the product during shipment and storage. The can shall be hermetically sealed under an atmosphere of nitrogen so that the oxygen content of the headspace gases in the filled and sealed container shall not exceed 2.0 percent when tested in accordance with 4.5.6. The can shall not leak when examined in accordance with the procedures in PPP-C-29.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1 except that cans with or without commercial exterior coating will be acceptable. Alternatively, cans may be made from 0.20 pound per base box electrolytic tinplate, provided the cans have an exterior commercial coating.

5.2 Packing. Twenty-four cans of product, of one type only arranged 4 in length, 3 in width, and 2 in depth, shall be packed on end in a snug-fitting shipping container in accordance with level A, B or C, as specified see 6.1).

5.2.1 Level A packing. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V2s of PPP-B-636. The tiered cans shall be separated with a full length and width fiberboard pad made of the same material as the box. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V3c, V3s, or V4s of PPP-B-636. The tiered cans shall be separated with a full length and width fiberboard pad made of the same material as the box. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. The shipping container shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except fiberboard containers shall be closed in accordance with Method II as specified in the appendix of PPP-B-636. The tiered cans shall be separated with a full length and width pad.

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5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling and marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. Labeling shall include the following information for type I:

CARROTS, DICED, DEHYDRATED, COMPRESSED
Net Weight
Name and address of supplier
Date Packaged
Preserved with Sulfur Dioxide
GAS-PACKAGED

DIRECTIONS FOR USE

Buttered carrots	50 servings
Carrots, dehydrated, compressed	16 oz. (1 #2 1/2 can)
Water, boiling	2 gallons
Salt	2-1/3 tbsp (1-1/2 oz.)
Butter, melted	8 oz. (1 cup)

Add carrots to boiling, salted water. Cover, return to boil and simmer until carrots separate from disc (approximately 3-5 minutes). Remove from heat. Let stand 10 minutes. Drain. Combine 1 cup drained liquid with butter. Pour over carrots. 16 ounces of dehydrated, compressed carrots equals approximately 11 pounds of cooked, trimmed fresh or frozen carrots or two No. 10 cans of canned carrots.

For type II:

CARROTS, SHREDDED, DEHYDRATED, COMPRESSED
Net Weight
Name and address of supplier
Date Packaged
Preserved with Sulfur Dioxide
GAS-PACKAGED

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DIRECTIONS FOR USE

Carrots, shredded, dehydrated, compressed	12 oz. (1 #2-1/2 can)
Water, cool (60°F)	1 gallon

Break each disc into four approximately equal portions. Add the portions to the water and allow to stand until fully rehydrated (approximately 45 minutes). Stir to break apart during rehydration. Drain. Substitute in any recipe using carrots, fresh shredded or ground.

12 oz. (1 #2-1/2 can) may be substituted for 6 (quarts) raw, shredded or ground carrots.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this document.
b. Type of product required.
c. When a first article is required (see 3.1, 4.4 and 6.4.1).
d. Provisions for approved preproduction samples (see 3.5.3 and 6.4).
e. Levels of preservation and packing required (see 5.1 and 5.2).
f. Type and class of unit load when unit loading is specified (see 5.3).
g. When type III exterior coating of cans is desired (see 5.1.1).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Level C packing. Packing specified in 5.2.3 is intended for direct shipment from the supply source to the first consuming activity within CONUS for immediate consumption.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.5 Supersession data. This document supersedes LP/P. DES 3-75A, Carrots, Dehydrated, Compressed, dated 21 February 1980.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8915-0776

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44116	Carrots, Dehydrated, Compressed, Diced or Shredded

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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